

Ultra Precision Motion Control of a Multiple Degrees of Freedom Magnetic Suspension Stage

Ximin Shan, Shih-Kang Kuo, Jihua Zhang, and Chia-Hsiang Menq

Abstract—A general framework for ultra precision motion control of magnetic suspension actuation systems with large travel ranges in multiple degrees of freedom (DOF) is presented in this paper. It encompasses the development of nonlinear electromagnetic force model for 6-DOF actuation, and the design of the necessary control architecture for ultra precision motion control of magnetic suspension actuation systems. A 6-DOF magnetic suspension stage (MSS) was designed and fabricated to illustrate the developed framework. The MSS consists of multiple electromagnets that are located around the flotor and are utilized to suspend and modulate its position and orientation. The control architecture takes the six control parameters provided by a laser measuring system and intends to control the 6-DOF motion through regulating the current in the electromagnets. The developed robust nonlinear control architecture consists of three components: 1) feedback linearization; 2) force distribution; and 3) H_∞ robust controllers for each DOF of motion. Several experiments are designed to illustrate the desired characteristics of the developed system in terms of stabilization, positioning, tracking, and contouring. The developed MSS can be used in applications requiring ultra precision motion control in multiple DOF.

Index Terms—Electromagnetic force modeling, feedback linearization, magnetic suspension, robust control, H_∞ synthesis, 2-DOF control.

I. INTRODUCTION

ADVANCED TOOL actuation technologies that deliver ultra precision positioning accuracy with high bandwidth are critically important to modern fabrication processes such as micro/nano fabrication and ultra precision machining. Following one of the most important principles in traditional precision engineering, current precision machinery are designed to operate with respect to a number of kinematic constraints, including linear guide ways and rotating axes, that ensure the necessary precision. These kinematic references are often assemblages of mechanical members and need be compounded or cascaded in order to achieve multiple degrees of freedom (DOF) motion, thus hindering the implementation of an actuation scheme with high bandwidth. In addition, the limitation of friction losses due to mechanical contacts is an engineering challenge when ultra precision actuation is desired. It is therefore very difficult to develop, based on current technology, high performance actuation systems capable of large

travel in multiple DOF with nanometer tracking accuracy at elevated bandwidths. As magnetic suspension is contact-free, there is virtually no friction and wear, thereby enabling ultra precision positioning accuracy, and multiple DOF actuation can be realized without mechanical compounding or cascading, thus providing high bandwidth precision tracking. Therefore, magnetic suspension is perceived to be a feasible alternative in the development of high performance actuation systems and is at the forefront of technologies resulting in technically advanced products that deliver superior performance with longer life.

Several magnetic suspension actuation systems have been developed by researchers. A magnetic levitation servo for flexible assembly was developed to provide 5-DOF motion with $1\text{-}\mu\text{m}$ resolution [18]. The servo was primarily utilized in flexible assembly as a programmable Remote Center Compliance device. The maximum travel of the floating body in translation and rotation were $\pm 0.6\text{ mm}$ and ± 0.7 degrees, respectively. An actuator for use as a probing tool of semiconductor devices was developed to provide positioning along two axes [2]. A positioning repeatability of $0.2\text{ }\mu\text{m}$ was obtained for travel lengths of about 1 mm . A 6-DOF magnetically levitated positioning stage was developed with vacuum compatibility [1], achieving a positioning noise below 100 nm peak-to-peak within a 0.3 mm^3 of travel. Moreover, a precision magnetic bearing stage has been developed for photolithography [15], [16], [19]. This stage could replace the mechanical bearing stages currently used in wafer steppers. The stage is suspended in 5-DOF by dc electromagnets, and driven in the sixth direction by a Lorentz-type linear motor. The stage achieves a positioning stability of 3 nm peak to peak in the bearing axes, and 15 nm in the axial direction. Although, the stage possesses a linear travel of 50 mm in the axial direction, the other 5-DOF operate in extremely narrow ranges. In another design, positioning accuracy of 0.1 nm inside a $100\text{-}\mu\text{m}$ cube has also been achieved [7], [8]. These investigations clearly demonstrate the viability of utilizing magnetic suspension for ultra fine motion control. However, motion ranges are small in either all or some directions in these systems.

Control algorithms for magnetic suspension systems have been extensively investigated in active magnetic bearing literature [13]. Magnetic suspension bearings are narrow gap devices that seek to maintain their nominal operating position, in spite of external disturbances. Linear controllers, based on a linearized model around the operating point, are usually utilized to regulate performance. Since the recent generation of magnetic suspension actuators has been built using electromagnets in push-pull configurations similar to those employed in magnetic bearings, researchers have pursued the same principles in the

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design of control systems for the magnetic suspension actuation systems. In these systems however, changing positions of the flotor corresponds to changing models of the electromagnetic force distribution. The linear controller performs satisfactorily in small regions around the nominal operating point, but the performance degrades with increasing travel. The requirement of large travel therefore necessitates the utilization of nonlinear control algorithms. Nonlinear control algorithms utilize the nonlinear model of the electromagnetic field distribution, describing significantly large regions around the electromagnet. Geometric feedback linearization techniques have been investigated for a 1-DOF magnetic suspension system [17], [10]. It was experimentally demonstrated that the controller yielded transient responses largely independent of the operating point air gap, when distances were increased four times in comparison to linear controllers. However, implementation of feedback linearization technique requires accurate system models. Otherwise controller performance degrades in the presence of uncertainties.

This paper presents a general framework for ultra precision motion control of magnetic suspension actuation systems with large travel ranges in multiple DOF. It encompasses the development of nonlinear electromagnetic force model for 6-DOF actuation, and the design of the necessary control architecture for ultra precision motion control of magnetic suspension actuation systems. In order to illustrate the developed framework, a 6-DOF magnetically suspension stage (MSS) was designed and fabricated. The MSS consists of multiple electromagnets that are located around the flotor and are utilized to suspend and modulate its position and orientation. The remainder of this paper is organized as follows. In Section II, an overview of the design of the magnetic suspension stage is presented. In Section III, modeling nonlinear electromagnetic interaction between the flotor and its surrounding electromagnets as well as the effects of the resulting magnetic forces on the flotor's dynamics is formulated. Section IV presents a robust nonlinear control architecture, which consists of three components: 1) feedback linearization; 2) force distribution; and 3) H_∞ robust controllers for each DOF of motion. In Section V, several experiments are designed to illustrate the desired characteristics of the developed system in terms of stabilization, positioning, tracking, and contouring.

II. MSS

A MSS using electromagnetic forces is designed and fabricated to realize ultra precision multiple DOF actuation. In contrast to the traditional stages that move along three orthogonal guide slides to perform three-dimensional motions, the magnetic suspension actuator moves freely along and rotates about any axis in the workspace. This characteristic is achieved by not including any mechanical constraint into the system and it makes the developed system very flexible and able to operate at high bandwidths. However, kinematic references have to come into the system to achieve the desired precision. In the developed system, instead of using mechanical references, laser measurement systems along with high performance feedback control are used to provide the needed kinematic references.

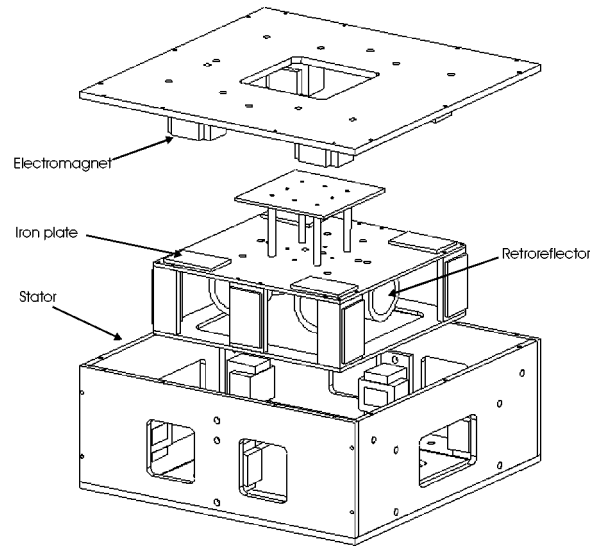


Fig. 1. Basic components of the MSS.

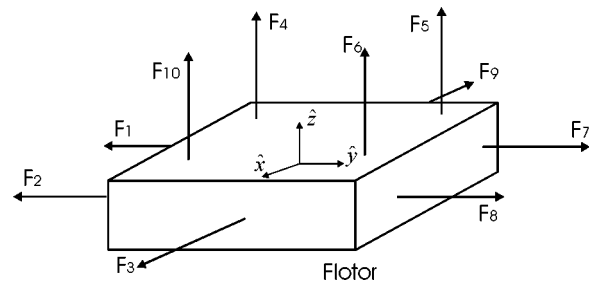


Fig. 2. Actuation scheme of the MSS.

A. Stage Design and Actuation Scheme

Fig. 1 shows an exploded view of the MSS with its basic components. In current implementation, the MSS consists of a flotor that can move freely within the work volume and is surrounded by ten dc electromagnets mounted on a stator. This configuration provides the capability of 6-DOF actuation. Six retro-reflectors are mounted on the flotor as part of the laser interferometer sensing system. The actuation scheme of the stage is illustrated in Fig. 2. The four electromagnetic actuators in vertical scheme provide the vertical suspension force. These four actuators also provide torques for rotations around x axis and y axis. The x axis actuation force is provided by two electromagnets along x axis. The rest four actuators in horizontal scheme provide the z axis torque in addition to the y axis actuation force.

B. Laser Interferometer Measurement System

A 6-DOF measurement system is developed and integrated with the MSS. The developed measurement system consists of a laser source and six interferometers with retroreflectors mounted on the flotor. The measurement system is based on six modified Michelson heterodyne interferometers [20]. Fig. 3 shows a prototype of the MSS equipped with the 6-DOF measurement system. Three laser beams are arranged along the vertical direction, and the other three are in the horizontal plane. This configuration is analytically and numerically verified to be

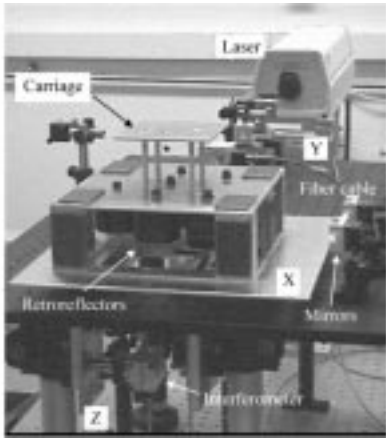


Fig. 3. A prototype of the MSA.



Fig. 4. A homemade interferometer.

free of singularity. Fig. 4 shows a prototype of the homemade interferometer. The resolution of the linear motion measurement of the homemade interferometer is 1.24 nm. By properly arranging the six retroreflectors and the six interferometers high rotational measurement sensitivity can also be achieved. Since the combination of a corner cube retroreflector and a plane mirror is used to replace the plane mirror in the homemade interferometer, the developed measuring system has a much larger rotational range compared to plane mirror laser interferometer systems. The measurement volume is dependent on the dimension of the assembly and on the size of each retroreflector.

C. Real-Time Measurement of 6-DOF Motions

The developed interferometer system consists of six interferometers. Using these interferometers with specific designs, the measured optical path change can be related to physical quantities such as linear and/or angular displacements along the transducer axes. In the developed system, these measured displacements need be integrated so as to compute the 6-DOF parameters of the controlled motion. The relation between the controlled 6-DOF motion and the six laser readings is derived [21]. Based on this relation, an advanced computational algorithm is developed to calculate in real time the six parameters of the resulting motion. Fig. 5 shows the output of the linear displacement in the z direction within a time period of 0.2 s. It is seen

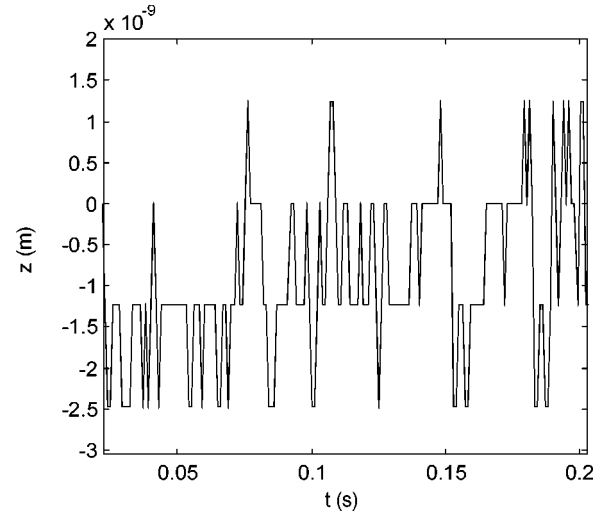


Fig. 5. Resolution test of the interferometer.

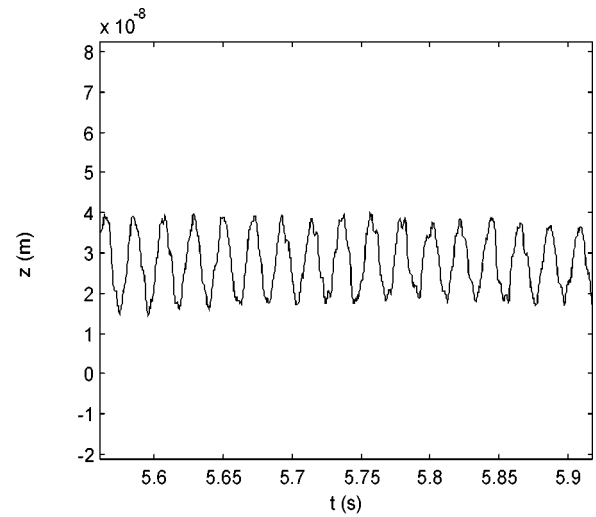


Fig. 6. Measured residual vibration.

that the developed measuring system has a resolution of 1.24 nm, and in a very short time, the influences of environment such as temperature change and the low frequency mechanical vibration are small. In order to examine the speed of the developed measuring system, a shock signal is applied to the stage, on which the retroreflectors are mounted. The six laser readings are processed in real time to determine the six parameters of the resulting vibration using the developed computational algorithm at a sampling frequency of 5 kHz. The stage initially has a few micrometers displacement in the z direction, and after 5.8 s, the residual vibration in the z axis is about 20 nm, as can be seen in Fig. 6. Similar results are observed for the other two linear motions and the three angular motions have amplitude about a few micro radians.

III. SYSTEM MODELING

The MSS consists of multiple electromagnets located around the flotor that are utilized to suspend and modulate its position and orientation. The electromagnetic interaction depends on the geometrical and material properties of the electromagnets, the

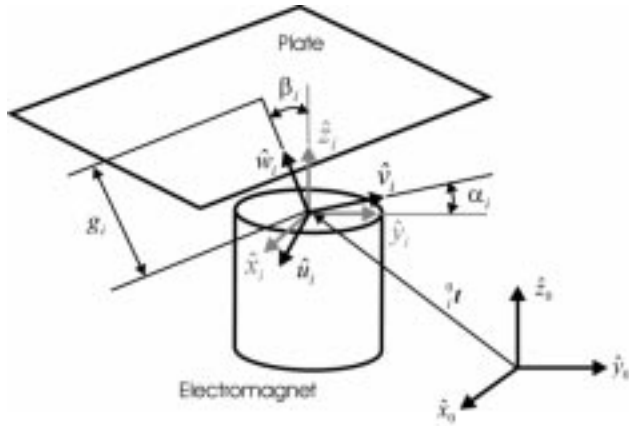


Fig. 7. Modeling electromagnetic force on inclined plates.

material properties of the flotor, its position and orientation relative to the electromagnets, and the current being supplied to the electromagnets. Since large travel is desired, modeling non-linear electromagnetic interaction between the flotor and its surrounding electromagnets as well as the effects of the resulting magnetic forces on the flotor's dynamics is necessary.

A. Nonlinear Electromagnetic Force Model

In deriving the electromagnetic force model for the MSS, it is assumed that the size of the flotor is significantly larger in comparison to the pole-faces of the electromagnets. Therefore, the interactions among the electromagnets in the MSS are minimal and can be ignored. Consequently, the force model for the complete MSS can then be derived by superimposing the interactions between the flotor and each individual electromagnet. The interaction between a single electromagnet and the flotor in turn can be effectively described by analyzing the magnetic forces between the electromagnet under consideration and the face of the flotor closest to it.

In order to obtain an expression for the forces that are exerted on the flotor due to the ten electromagnets, a reference coordinate system $[\hat{x}_o \ \hat{y}_o \ \hat{z}_o]$ and ten local coordinate systems $[\hat{x}_i \ \hat{y}_i \ \hat{z}_i]$, $i = 1, 2, \dots, 10$, are established as shown in Fig. 7. Each local coordinate system is attached to an electromagnet with $\hat{z}_i$ normal to and origin located at the center of its pole-face. The transformation between a local coordinate system and the reference coordinate system can be expressed as

$$[\hat{x}_i \ \hat{y}_i \ \hat{z}_i] = [\hat{x}_o \ \hat{y}_o \ \hat{z}_o] {}^o_i \mathbf{R} \quad (1)$$

where ${}^o_i \mathbf{R} = [{}^o_i \hat{x}_i \ {}^o_i \hat{y}_i \ {}^o_i \hat{z}_i]$ is the associated rotation matrix. The rotation matrix along with a translation vector ${}^o_i \mathbf{t}$ defines the position and orientation of the corresponding electromagnet.

1) *Magnetic Forces on an Inclined Plate:* The interaction between an electromagnet and the face closest to it corresponds to that of a metallic plate immersed in a magnetic field due to the electromagnet [9]. Since it is assumed that the size of the flotor is significantly larger in comparison to the pole-faces of the electromagnets, for simplicity the metallic plate under consideration can be treated as an infinite plate, and thus its position and orientation relative to the electromagnet can be characterized by a gap, g_i , and by two angles, α_i and β_i , respectively.

The first angle α_i is the in-plane rotation angle about $\hat{z}_i$, and the second angle β_i is the plate inclination angle about the rotated $\hat{y}_i$. With these two angles, a new coordinate system can be established as follows:

$$[\hat{u}_i \ \hat{v}_i \ \hat{w}_i] = [\hat{x}_i \ \hat{y}_i \ \hat{z}_i] \mathbf{R}_z(\alpha_i) \mathbf{R}_y(\beta_i). \quad (2)$$

In (2), " $\mathbf{R}_k(\theta)$ " is a rotational operator which performs a rotation about the axis direction $\hat{k}$ by an angle θ [3], and the resulting $\hat{w}_i$ is normal to the plate.

The electromagnetic forces, which are exerted on the plate, are modeled as equivalent force $[f_{\hat{u}_i} \ f_{\hat{v}_i} \ f_{\hat{w}_i}]^T$ and torque $[\tau_{\hat{u}_i} \ \tau_{\hat{v}_i} \ \tau_{\hat{w}_i}]^T$ and are functions of the gap g_i , orientation angles α_i and β_i , and electromagnet current I_i . As the electromagnetic forces are linear with respect to the square of the current I_i^2 [9], [11], the electromagnetic forces and torques are expressed as follows:

$$\begin{aligned} f_{\hat{u}_i} &= P_{\hat{u}_i}(g_i, \alpha_i, \beta_i) I_i^2 \\ f_{\hat{v}_i} &= P_{\hat{v}_i}(g_i, \alpha_i, \beta_i) I_i^2 \\ f_{\hat{w}_i} &= P_{\hat{w}_i}(g_i, \alpha_i, \beta_i) I_i^2 \end{aligned} \quad (3)$$

$$\begin{aligned} \tau_{\hat{u}_i} &= Q_{\hat{u}_i}(g_i, \alpha_i, \beta_i) I_i^2 \\ \tau_{\hat{v}_i} &= Q_{\hat{v}_i}(g_i, \alpha_i, \beta_i) I_i^2 \\ \tau_{\hat{w}_i} &= Q_{\hat{w}_i}(g_i, \alpha_i, \beta_i) I_i^2 \end{aligned} \quad (4)$$

where P and Q are nonlinear functions in terms of g_i , α_i , and β_i . These functions can either be determined by performing three dimensional electromagnetic finite element simulations [9], or by experimental calibrations.

In the case of an axially symmetric electromagnet, P and Q are independent of the in-plane rotation angle α_i . Therefore, P and Q need to be calibrated only with respect to the gap g_i and the inclination angle β_i . In addition, due to the symmetry in the interaction between the electromagnet and the infinite plate, force component $f_{\hat{v}_i}$ and torque components $\tau_{\hat{u}_i}$, $\tau_{\hat{w}_i}$ are identically zero for all values of g_i and β_i . Consequently, the remaining components of the electromagnetic forces and torques are expressed as follows:

$$\begin{aligned} f_{\hat{u}_i} &= P_{\hat{u}_i}(g_i, \beta_i) I_i^2 \\ f_{\hat{w}_i} &= P_{\hat{w}_i}(g_i, \beta_i) I_i^2 \\ \tau_{\hat{v}_i} &= Q_{\hat{v}_i}(g_i, \beta_i) I_i^2. \end{aligned} \quad (5)$$

It is evident that if axially symmetric electromagnets are employed modeling electromagnetic force is much simpler.

In the case of 3-DOF linear motion control, the system can be designed in a way such that each plate of the flotor is always parallel to the pole-face of the corresponding electromagnet. Consequently, the plate experiences a normal force $f_{\hat{w}_i}$, with the other five components being identically zero. The electromagnets utilized in the current MSS were calibrated using a force/torque sensor along with a calibration fixture developed in laboratory. The geometry and physical properties of the E-shape electromagnets are given in Table I. The cores of the electromagnets are made of silicon steel laminations. In the experiment, the carbon steel plate was always parallel to the pole-face of the electromagnet and the air gap was adjusted to distinct

TABLE I
PHYSICAL CHARACTERISTICS OF E-SHAPE ELECTROMAGNET

Physical characteristic	Value
Diameter of winding wire	0.0375 inch
Number of winding turns	115
Resistance of wire	0.6 ohm
Length of electromagnet	1.875 inch
Width of electromagnet	1.0 inch
Height of electromagnet	1.25 inch
Cross-sectional area of electromagnet	1.25 inch ²

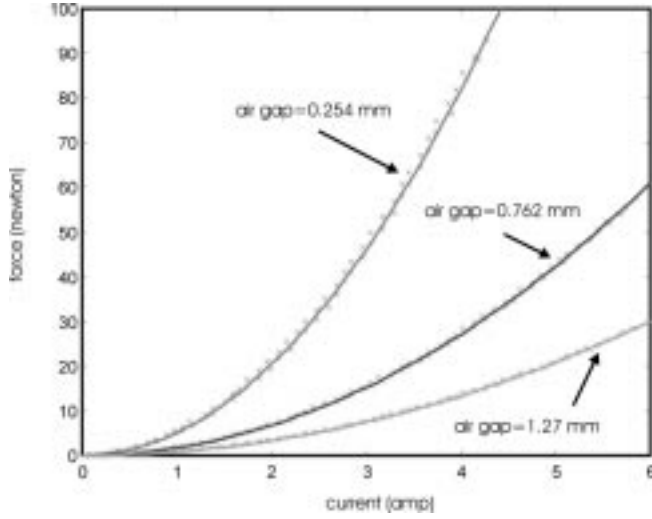


Fig. 8. Calibration data and electromagnetic force model.

values. Fig. 8 shows some of the experimental data and the corresponding curve fitting results. The fitted curve is characterized by the following relationship:

$$f_{\hat{w}_i} = C_i \frac{I_i^2}{(\alpha_i + g_i)^2}. \quad (6)$$

In the equation, g_i denotes the measured air gap, I_i is the coil current, and C_i and α_i are two constant parameters. The calibrated values for this electromagnet are $C_i = 2.418 \times 10^{-6} \text{ N} \cdot \text{m}^2/\text{A}^2$, and $\alpha_i = 4.335 \times 10^{-4} \text{ m}$.

2) *Net Magnetic Forces on the Flotor*: The net magnetic forces on the flotor can be derived by superimposing the interactions between each individual electromagnet and their corresponding plates. For this purpose, the magnetic forces derived in the previous section need to be referred to a common reference coordinate system. Moreover, in order to analyze the magnetic forces between the electromagnet under consideration and the plate closest to it, the plate's position and orientation need to be related to the motion of the flotor. Therefore, a moving coordinate system $[\hat{x}_f(t) \ \hat{y}_f(t) \ \hat{z}_f(t)]$ is established and attached to the mass center of the flotor as shown in Fig. 9. The plate's position and orientation are characterized by a normal vector $\hat{w}_{fi}$ along with the shortest distance ℓ_{fi} from the origin of this moving coordinate system. A coordinate system attached to the plate can be established according to $\hat{w}_{fi}$ and can be expressed with respect to the moving coordinate system as follows:

$$[\hat{u}_{fi} \ \hat{v}_{fi} \ \hat{w}_{fi}] = [\hat{x}_f \ \hat{y}_f \ \hat{z}_f] \mathbf{R}_z(\alpha_{fi}) \mathbf{R}_y(\beta_{fi}). \quad (7)$$

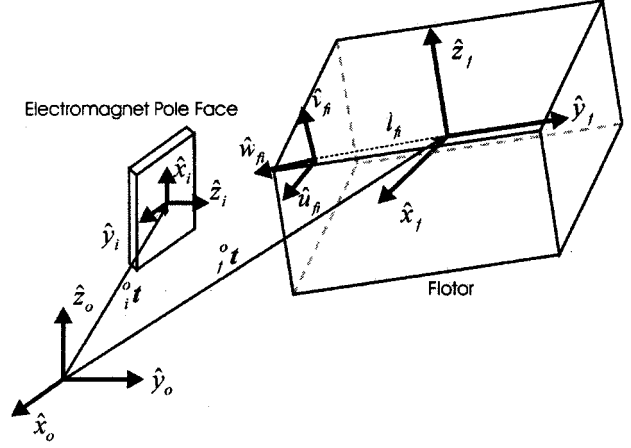


Fig. 9. Modeling net electromagnetic force on the flotor.

When the flotor moves, the position and orientation of the flotor, and thus each plate, vary with time. With the 6-DOF motion represented by a rotation matrix ${}^o_f\mathbf{R}(t)$ and a translation vector ${}^o_f\mathbf{t}(t)$, the time variant orientation and position of each plate are derived as follows:

$${}^o\mathbf{w}_{fi}(t) = {}^o_f\mathbf{R}(t) {}^f\mathbf{w}_{fi}, \quad (8)$$

$${}^o\mathbf{t}_{fi}(t) = {}^o_f\mathbf{t}(t) + \ell_{fi} {}^o\mathbf{w}_{fi}(t). \quad (9)$$

With (8), the plate inclination angle and in-plane rotation angle can then be derived such that $\hat{w}_i(t) = -\hat{w}_{fi}(t)$

$$\beta_i(t) = \cos^{-1}[-({}^f\mathbf{w}_{fi}^T {}^o_f\mathbf{R}^T(t) {}^o\mathbf{z}_i)] \quad (10)$$

in which $0 \leq \beta_i(t) < \pi/2$. If $\beta_i(t) \neq 0$

$$\alpha_i(t) = \tan^{-1} \left[\frac{-({}^f\mathbf{w}_{fi}^T {}^o_f\mathbf{R}^T(t) {}^o\mathbf{y}_i)}{-({}^f\mathbf{w}_{fi}^T {}^o_f\mathbf{R}^T(t) {}^o\mathbf{x}_i)} \right]. \quad (11)$$

If $\beta_i(t) = 0$, (11) is singular, and $\alpha_i(t)$ is set to zero. The gap is defined as the shortest distance from the pole-face center of the electromagnet to the plate. It is the distance from the pole-face center to the associated foot point on the plate. Consequently, the gap can be calculated as follows:

$$g_i(t) = {}^f\mathbf{w}_{fi}^T {}^o_f\mathbf{R}^T(t) [{}^o\mathbf{t} - {}^o_f\mathbf{t}(t)] - \ell_{fi}. \quad (12)$$

With the force models derived in the coordinate $[\hat{u}_i \ \hat{v}_i \ \hat{w}_i]$, as given in (3) and (4), the net electromagnetic forces and torques exerted on the flotor can be derived using superposition. The net forces and torques are acting on the flotor at the center of mass and can be expressed along the three reference axes. Since

$$[{}^o\mathbf{u}_i(t) \ {}^o\mathbf{v}_i(t) \ {}^o\mathbf{w}_i(t)] = {}^o_i\mathbf{R}\mathbf{R}_z(\alpha_i(t))\mathbf{R}_y(\beta_i(t)) \quad (13)$$

the net electromagnetic forces and torques can be derived as

A. Feedback Linearization and Current Distribution for 6-DOF Motion Actuation

For a nonlinear system, linear controllers based on a linearized model at an operating point are only effective in a small neighborhood around that point. Out of this range the performance of the system often rapidly degrades. The approach based on feedback linearization can algebraically transform a nonlinear system dynamics into a linear one, based on which linear control design methods can be applied [14]. The rotational motion dynamics expressed in (21) is nonlinear in terms of $\mathbf{q}$ and $\dot{\mathbf{q}}$. The feedback linearization process for torque compensation can be formulated by defining the right hand side of (21) as the virtual control effort, τ'

$$\tau' = \mathbf{I}(\mathbf{q})^{-1}(\tau - \mathbf{V}(\mathbf{q}, \dot{\mathbf{q}})). \quad (22)$$

Then the rotational motion is characterized by

$$\ddot{\mathbf{q}} = \tau'. \quad (23)$$

And the compensated torque τ can be calculated as follows:

$$\tau = \mathbf{I}(\mathbf{q})\tau' + \mathbf{V}(\mathbf{q}, \dot{\mathbf{q}}). \quad (24)$$

Straightforwardly, the resulting system is linear with respect to the new control variable $\mathbf{f}$ and τ' . On the other hand, the net forces $\mathbf{f}$ and torques τ are to be generated by currents in the electromagnets. Based on the expression of net electromagnetic forces and torques shown in (14) and (15), just for notational convenience, the nonlinear relation between the currents and the resulting forces and torques is rewritten as

$$\begin{bmatrix} \mathbf{f} \\ \tau \end{bmatrix} = \mathbf{G}(\mathbf{f}, \mathbf{t}, \mathbf{q}) \begin{bmatrix} I_1^2 \\ \vdots \\ I_{10}^2 \end{bmatrix} \quad (25)$$

where $\mathbf{G}(\mathbf{f}, \mathbf{t}, \mathbf{q})$ is a 6×10 nonlinear matrix that can be determined by the locations of the electromagnets and the position and orientation of the flotor as shown in (13) and (14). Finding physically realizable solution of (25) is a *current distribution* problem in the control architecture. When properly designed, the matrix $\mathbf{G}$ in (25) is of full row rank. Therefore, all solutions of the current distribution can be expressed as follows:

$$\begin{bmatrix} I_1^2 \\ \vdots \\ I_{10}^2 \end{bmatrix} = \mathbf{G}^T(\mathbf{G}\mathbf{G}^T)^{-1} \begin{bmatrix} \mathbf{f} \\ \tau \end{bmatrix} + \sum_{k=1}^4 \lambda_k \mathbf{v}_k \quad (26)$$

where $\mathbf{v}_k, k = 1, \dots, 4$, form a basis of the null space of matrix $\mathbf{G}$. In order to yield a realizable current distribution, parameters λ_k must be appropriately selected to ensure the positiveness of the solution. For optimal conditioning of the null space current vectors, readers are referred to [9]. Together with (24), this equation describes a nonlinear mapping from the virtual control efforts, i.e., $\mathbf{f}$ and τ' , of a linearized plant model, to the currents that can be commanded physically. However, due to modeling uncertainties in (25), perfect linearization is impossible and the derived linear plant model is only an approximation to the compensated system. Nevertheless, the effect of this discrepancy can be much reduced after employing an additional robust

linear controller, the design of which is based on an approximate plant model. The approximate plant model can be either the derived linear plant model or an experimental plant model obtained through system identification.

In the case of 3-DOF linear motion actuation of the magnetic suspension stage, the nonlinearity caused by rotational motion is negligible. Particularly, the Coriolis and centrifugal term $\mathbf{V}(\mathbf{q}, \dot{\mathbf{q}})$ is usually much smaller when compared to $\mathbf{I}(\mathbf{q})\tau'$. Therefore, implementation of (24) can be partially or fully neglected for linear motion control.

B. Force Distribution for 3-DOF Linear Motion Actuation

For the case of 3-DOF linear motion actuation of the magnetic suspension stage, the current distribution problem can be simplified by dividing it into two parts: force distribution and current-force feedback linearization that uses a simplified force model as shown in (6). For the developed magnetic suspension stage, the force distribution problem can be solved by dividing the electromagnets into two groups, i.e., the horizontal group that generates F_1, F_2, F_3, F_7, F_8 , and F_9 , and the vertical group that generates F_4, F_5, F_6 , and F_{10} , as shown in Fig. 2. For the horizontal scheme, three intermediate force variables are defined as follows: $F_{17} = F_7 - F_1, F_{28} = F_8 - F_2$, and $F_{93} = F_3 - F_9$. The relationship between the net forces and torques and the electromagnetic actuation forces in the horizontal scheme can be expressed as

$$\begin{bmatrix} f_{\hat{x}_o}^n \\ f_{\hat{y}_o}^n \\ \tau_{\hat{z}_o}^n \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ -\ell_{17} & \ell_{28} & 0 \end{bmatrix} \begin{bmatrix} F_{17} \\ F_{28} \\ F_{93} \end{bmatrix} = \mathbf{G}_1 \cdot \begin{bmatrix} F_{17} \\ F_{28} \\ F_{93} \end{bmatrix} \quad (27)$$

where ℓ_{17} and ℓ_{28} are moment arms of force F_{17} and F_{28} , respectively. Obviously, the intermediate force variables can be obtained from the desired net forces and torque by inverting $\mathbf{G}_1$. However, it is also obvious that the choice of horizontal electromagnetic actuation forces is not unique. To keep the operation of the stage efficient on power consumption, the electromagnet pairs that generate F_{17}, F_{28} , and F_{93} can work in a so-called “push-pull” configuration. For example, positive F_{93} is distributed to F_3 only while negative F_{93} to F_9 only. That is

$$F_3 = \begin{cases} F_{93}, & \text{if } F_{93} > 0 \\ 0, & \text{otherwise} \end{cases} \quad (28)$$

and

$$F_9 = \begin{cases} -F_{93}, & \text{if } F_{93} < 0 \\ 0, & \text{otherwise.} \end{cases} \quad (29)$$

Similarly, for vertical scheme the transform from actuation forces to net forces is

$$\begin{bmatrix} f_{\hat{z}_o}^n \\ \tau_{\hat{x}_o}^n \\ \tau_{\hat{y}_o}^n \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -\ell_{4x} & \ell_{5x} & \ell_{6x} & -\ell_{10x} \\ \ell_{4y} & \ell_{5y} & -\ell_{6y} & -\ell_{10y} \end{bmatrix} \begin{bmatrix} F_4 \\ F_5 \\ F_6 \\ F_{10} \end{bmatrix} = \mathbf{G}_2 \cdot \begin{bmatrix} F_4 \\ F_5 \\ F_6 \\ F_{10} \end{bmatrix} \quad (30)$$

where ℓ_{kx} and $\ell_{ky}, k = 4, 5, 6, 10$, are the moment arms of k th electromagnet with respect to x axis and y axis, respectively. It is obvious that the solution to (30) for the vertical scheme force

distribution is not unique either. For the current developed MSS, the pseudo-inverse transform is adopted to calculate the vertical force distribution

$$\begin{bmatrix} F_4 \\ F_5 \\ F_6 \\ F_{10} \end{bmatrix} = G_2^T (G_2 G_2^T)^{-1} \begin{bmatrix} f_{\ddot{z}_o}^n \\ \tau_{\ddot{x}_o}^n \\ \tau_{\ddot{y}_o}^n \end{bmatrix}. \quad (31)$$

With this approach, the energy consumption in vertical scheme is also minimized. In the actuation scheme shown in Fig. 2, the changes of moment arms due to translation motion of the flotor are small. Therefore, ℓ_{kx} and ℓ_{ky} are always greater than their minima, $\ell_{kx} \geq \ell_x$ and $\ell_{ky} \geq \ell_y$, where ℓ_x and ℓ_y are two constant and positive values. The pseudo-inverse solution can be guaranteed to be positive if the following condition is satisfied

$$\left| \frac{\tau_{\ddot{x}_o}^n}{\ell_x} \right| + \left| \frac{\tau_{\ddot{y}_o}^n}{\ell_y} \right| < f_{\ddot{z}_o}^n = m(g + \ddot{z}). \quad (32)$$

For 3-DOF linear actuation, $|\tau_{\ddot{x}_o}^n|$ and $|\tau_{\ddot{y}_o}^n|$ are small, therefore, this condition can be easily satisfied if $|\ddot{z}| \ll g$.

The current-force feedback linearization followed by force distribution can be formulated from (6) as shown below

$$I_i^2 = \frac{F_i}{C_i} (a_i + g_i(\ddot{f}t))^2, \quad i = 1, 2, \dots, 10. \quad (33)$$

C. H_∞ Robust Controller Design

In addition to feedback linearization and force/current distribution described above, linear controllers need to be designed for 6-DOF motion control. If a perfect description of nonlinearities of the system were available, the linearized plant model would have consisted of six decoupled double integrators. Linear controllers could then be synthesized easily with either conventional pole placement approach, or linear quadratic (LQ) approaches when considering the influence of white noise to the performance of the system. However, modeling uncertainties are inevitable and perfect modeling is impossible. Moreover, a magnetic suspension system also suffers from parameter variations such as magnetization drifting [6], [9] and external disturbances that need be effectively rejected for precision motion control. A robust controller is thus needed so as to overcome modeling uncertainties and to reject external disturbances. In this research, the H_∞ robust control synthesis technique [4] is adopted.

The H_∞ controllers for the MSS system can be designed by using either the derived nominal plant model or an experimental model through system identification. Unlike stable systems, identification data of the MSS system cannot be obtained from open loop response. Therefore, lead-lag compensators are initially adopted for preliminary closed-loop performance investigation and for system identification. In system identification, periodic position reference signals containing multiple sinusoidal components are commanded to the closed-loop system, and the resulting position output and virtual control effort (force or torque) are recorded for analysis with a sampling rate of 1000 samples per second. Conventional iterative least square fitting technique is then applied to yield a parametric model of the com-

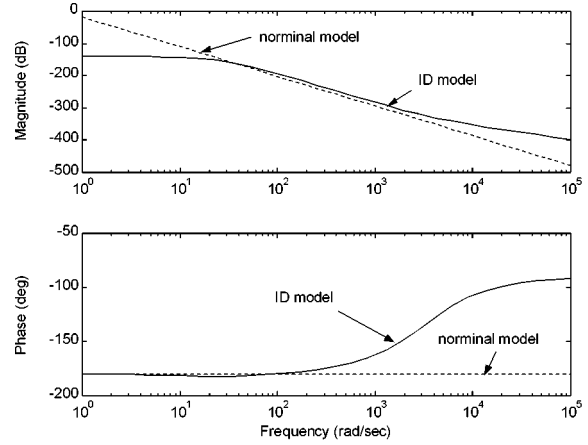


Fig. 11. Frequency response of x axis motion.

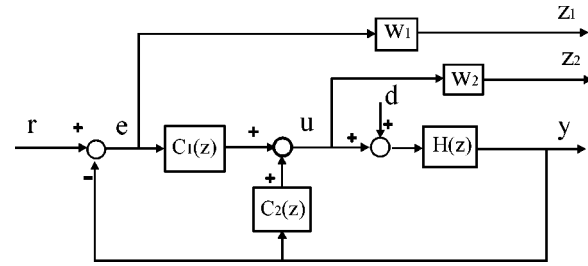


Fig. 12. A 2-DOF H_∞ controller for the MSS.

pensated system from the experimental data. To illustrate the design process, the identified system model along x axis and synthesis of H_∞ controller for it are presented in this paper. Similar design procedures are employed when designing controllers for other axes. A second order discrete-time model for the translation motion along x axis is obtained from the experimental data

$$H(z) = \frac{10^{-6} \times (0.5637z + 0.1246)}{z^2 - 2.0034z + 1.0026}. \quad (34)$$

By comparing the frequency responses of both the identified model and the derived double integrator, as shown in Fig. 11, the discrepancy at both low and high frequencies is evident.

For unstable plants with conventional 1-DOF control, the closed-loop performance is usually characterized by a large overshoot in step response [5]. To reduce overshoot in ultra precision positioning and tracking, 2-DOF feedback control is adopted for the MSS. Fig. 12 depicts the 2-DOF feedback control diagram, where w_1 is the weighting function of tracking error e , and w_2 is the weighting function of control effort u . Minimizing z_1 can ensure small tracking error within desired bandwidth, while minimizing z_2 can avoid actuator saturation as well as implicitly limiting the bandwidth of the closed-loop system to avoid amplifying measurement noise and exciting unmodeled dynamics. The augmented plant shown in Fig. 12 can be presented as follows:

$$\begin{bmatrix} z_1 \\ z_2 \\ e \\ y \end{bmatrix} = \begin{bmatrix} w_1(z) & -w_1(z)H(z) & -w_1(z)H(z) \\ 0 & 0 & w_2(z) \\ I & -H(z) & -H(z) \\ 0 & H(z) & H(z) \end{bmatrix} \begin{bmatrix} r \\ d \\ u \end{bmatrix}. \quad (35)$$

The compensators $C_1(z)$ and $C_2(z)$ in the 2-DOF control scheme shown in Fig. 12 can be synthesized in state space H_∞ framework [4], [22]. The computational burden can be greatly relieved by using MATLAB computational environment with Robust Control Toolbox. Therefore, major effort can be concentrated on how to choose appropriate weighting functions. To ensure zero steady-state error for ultra precision positioning, the closed-loop system should be at least type I, while the system needs be type II if zero tracking error to a ramp reference input is further required. This can be realized by introducing a number of integrators in the weighting function w_1 . Two types of controllers, namely positioning controllers and tracking controllers, are synthesized and implemented in this research in order to meet the performance requirements for different applications.

In general, however, H_∞ design framework does not produce integral control. There are several ways to achieve the integral design [22]. In this research, the approximated integral control scheme is adopted by introducing a small perturbation in the integrators of the weighting function w_1 . The weighting function w_1 of the positioning controller is designed as follows:

$$w_1(z) = \frac{\alpha}{z - 1 + \varepsilon}. \quad (36)$$

And for the tracking controller, it is

$$w_1(z) = \frac{\beta(z - 0.985)(z - 0.98)}{(z - 1 + \varepsilon)^2}. \quad (37)$$

In these two equations, α and β are constants that can be tuned for each axis, and $\varepsilon = 10^{-6}$. The weighting function w_2 is designed to be a constant for each axis. For x axis positioning, the parameters are selected to be $\alpha = 0.03$ and $w_2 = 10^{-6}$ and the resulting controllers are given below

$$C_1(z) = \frac{1.1622 \times 10^4(z^3 + 1.2242z^2 + 0.2217z)}{z^3 - 0.5352z^2 - 0.4109z - 0.0539} \quad (38)$$

$$C_2(z) = \frac{-5.4094 \times 10^5(z^2 + 0.1983z - 0.8017)}{z^2 + 0.4648z + 0.0539}. \quad (39)$$

For x axis tracking, the parameters are selected to be $\beta = 0.0003$ and $w_2 = 4 \times 10^{-10}$. And the resulting controllers are in

$$C_1(z) = \frac{-2.7259 \times 10^4(z^4 - 2.3605z^3 + 1.0061z^2 + 0.3484z)}{z^4 - 1.3695z^3 - 0.1705z^2 + 0.4495z + 0.0905} \quad (40)$$

$$C_2(z) = \frac{-1.0388 \times 10^6(z^2 + 0.2988z - 0.7012)}{z^2 + 0.6305z + 0.0905}. \quad (41)$$

V. EXPERIMENTAL RESULTS

The traveling volume of the current MSS is $4 \times 4 \times 2 \text{ mm}^3$ in the 3-linear DOF, and is more than 1° angular displacement in each of the remaining three rotational DOF. The limitation on angular range is due to the geometric design of current electromagnetic actuation scheme. With an improved design, this traveling range can be increased. Nevertheless, in the manufacture

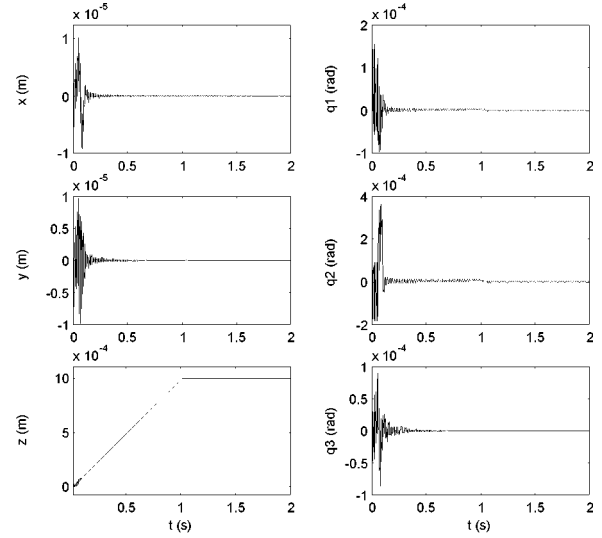


Fig. 13. A levitation process.

of semiconductor devices, the rotational motion can be used to compensate for the orientation placement error that is normally less than 1° . The nonlinear control architecture is implemented in discrete-time with a sampling rate of 1000 Hz. The mass of the flotor is $m = 2.4 \text{ Kg}$, and the principal moment of inertia are $I_x = 0.01137 \text{ Kg} \cdot \text{m}^2$, $I_y = 0.01179 \text{ Kg} \cdot \text{m}^2$, and $I_z = 0.01989 \text{ Kg} \cdot \text{m}^2$. Through force/current distribution and feedback linearization, the multiple-input-multiple-output (MIMO) system is decomposed to six single-input-single-output (SISO) subsystems. For each subsystem, H_∞ positioning and tracking controllers are synthesized based on identified plant models. A number of experiments were designed to verify the control algorithm and investigate the performance of the overall system, including stabilization, step response, tracking, and circular contouring.

A. Stabilization

Fig. 13 depicts the levitation and stabilization of the stage. In the experiment, the flotor rested on the stator initially and was levitated in the vertical direction to a height of 1 mm in 1 s. The left bottom plot shows the z axis motion trajectory, while other plots show the accompanying motions in the other 5-DOF, two translations (x and y) and three rotations (q_1 , q_2 , and q_3). A ramp trajectory, instead of a step signal, was used as the command input to the control system so as to have smoother levitation. After some transient oscillations the system was stabilized at the desired position. It is seen that the stage can be stabilized using current control algorithms. The steady-state positioning noise after stabilization is shown in Fig. 14. In this figure, positioning noises in all the three translational directions are within about $\pm 10 \text{ nm}$. The best angular positioning is achieved in q_3 , where the peak-to-peak positioning noise is smaller than 10^{-7} radian. The steady-state noises in rms are calculated in a period of 5 s and are listed in Table II. Fig. 15 shows the frequency spectrum of the noise in y axis positioning, analyzed using Fast Fourier Transform (FFT) on the 5000 points of data. There is a sharp spike at about 60 Hz.

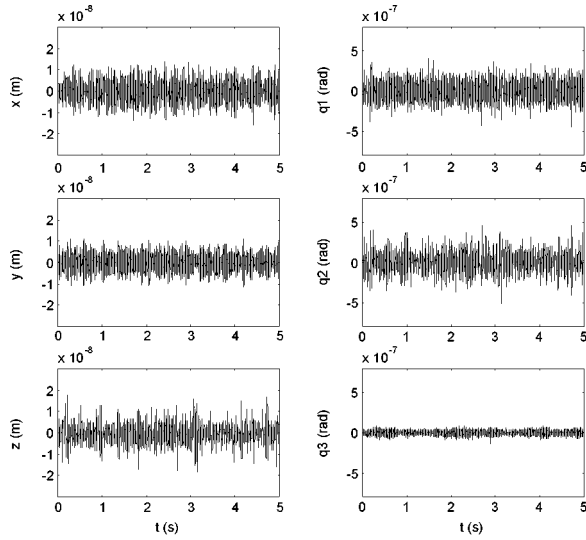


Fig. 14. Steady-state positioning noise.

TABLE II
STEADY-STATE POSITIONING NOISE (RMS)

Direction	Noise
x (nm)	4.3
y (nm)	3.8
z (nm)	4.7
q ₁ (μrad)	0.136
q ₂ (μrad)	0.132
q ₃ (μrad)	0.030

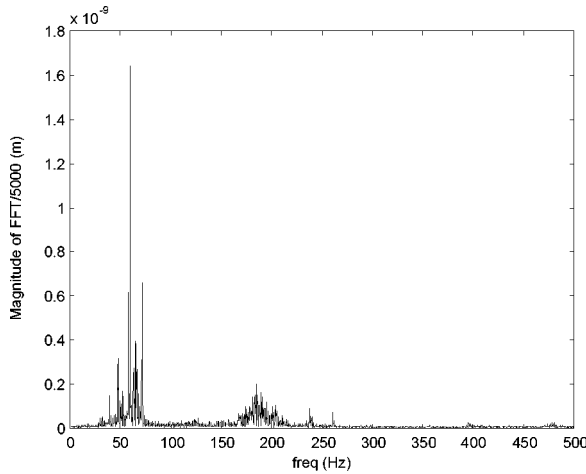
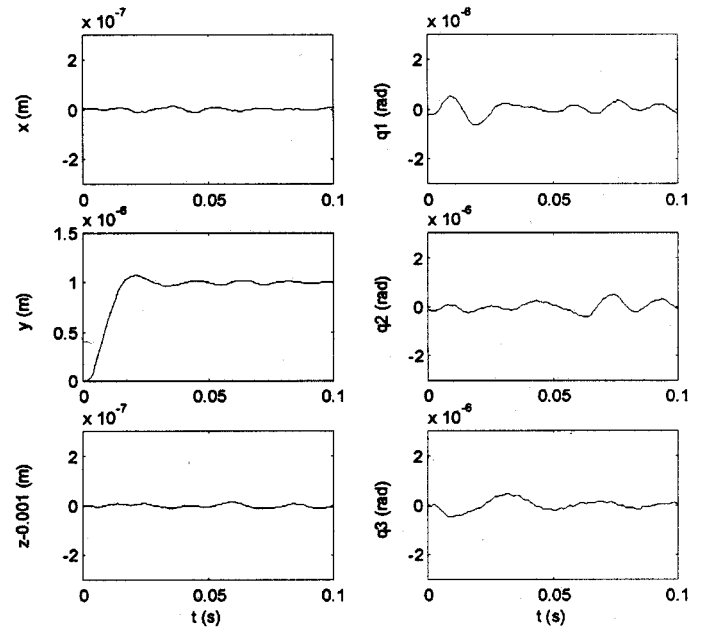
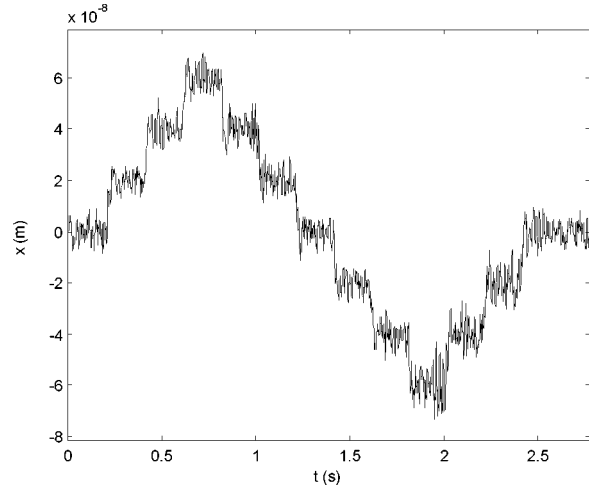


Fig. 15. FFT of x axis noise.

B. Positioning

The response to a 1 μm step input in y direction is shown in Fig. 16. In this experiment, the flotor was commanded to translate 1 μm in the positive y direction. The accompanying motions in other directions are also shown. Fast response and small accompanying motions can be seen in this figure. The peak time of the step response in this experiment is about 20 ms, and the overshoot is extremely small (about 7.5%). To further investigate the positioning resolution, the experimental result with multiple 20-nm steps in x direction is shown in Fig. 17.

Fig. 16. 1- μm step in translation in y direction.Fig. 17. 20-nm steps in x direction.

The steps can be clearly seen in this figure. Similar positioning performance can be obtained along other directions.

C. Tracking and Contouring

In order to illustrate the tracking and contouring abilities of the MSS, two operations were experimentally evaluated. In the first experiment, the stage was commanded to track a triangular wave along x direction. The amplitude of the triangular wave in this experiment is 1 mm. The tracking performance is illustrated in Fig. 18. It is seen that the stage precisely followed the command input until it was requested to make a turn. The steady-state tracking error is zero, while the position noise (not shown in the figure) is within ± 10 nm. In the second experiment, the stage was commanded to contour a circle in xy plane with a diameter of 0.6 mm in 8 s. The resulting trajectory of the stage along with the associated contouring error and tracking error is shown in Fig. 19. It is seen that the stage precisely tracks

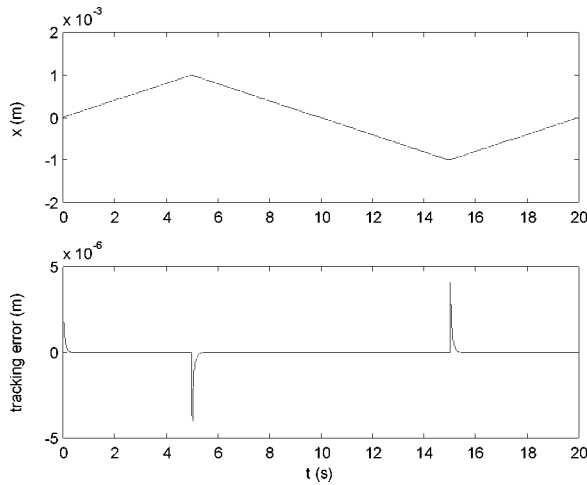


Fig. 18. X axis tracking control.

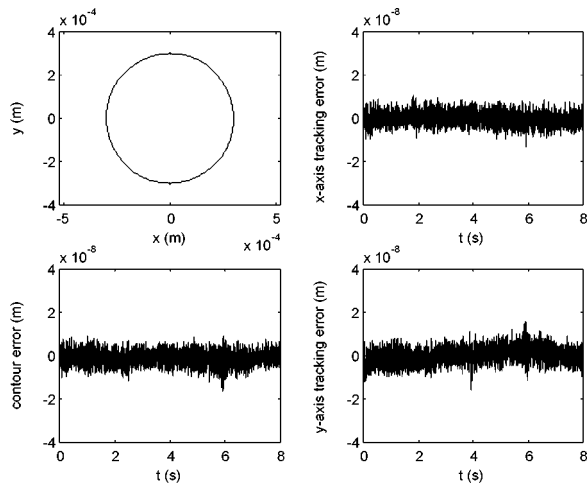


Fig. 19. Circular contouring control.

the circle. The contouring error and the tracking errors along both axes are within about ± 10 nm.

VI. CONCLUSIONS

In this paper, modeling nonlinear electromagnetic interaction between the flotor and its surrounding electromagnets is formulated. The resulting magnetic forces and their effects on the flotor's dynamics are expressed in terms of the 6-DOF motion of the flotor. This relation along with the developed nonlinear electromagnetic force model forms the basis of designing the necessary control architecture for ultra precision motion control of magnetic suspension actuation systems with large travel ranges in multiple DOF. In order to realize large travel motion control, a nonlinear control algorithm based on geometric feedback linearization theory is developed. Based on the resulting decoupled linear models, a 2-DOF control structure is adopted in the H_∞ synthesis framework so as to overcome modeling uncertainties and suppress external disturbances. A 3-DOF magnetically suspension stage (MSS) was designed and fabricated to illustrate the developed control algorithms. The MSS consists of multiple electromagnets that are located around the flotor and are utilized to suspend and modulate its position and orientation. The control architecture takes the six control parameters

provided by a laser measuring system and intends to control the 6-DOF motion through regulating the current in the electromagnets. Several experiments are designed to illustrate the desired characteristics of the developed system in terms of stabilization, positioning, tracking, and contouring.

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